

Q.22 Describe impact of silica content on viscosity of glass.

### SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the parts and working of a glass tank furnace with diagram.

Q.24 Describe the container glass manufacturing process with flowchart.

Q.25 Write various types of furnaces used in glass industry. Explain any one.

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220442

**4th Sem. / Ceramic Engineering**

**Subject : Glass Technology-1**

Time : 3 Hrs.

M.M. : 60

### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which oxide gives blue color to glass?

- a) Chromium oxide      b) Iron oxide
- c) Cobalt oxide        d) Zinc oxide

Q.2 Broken glass pieces used in tank furnace are known as:

- a) Grog                      b) Wood
- c) Cullet                    d) Pallet

Q.3 Which of the following is a network former?

- a)  $\text{Na}_2\text{O}$                       b)  $\text{SiO}_2$
- c)  $\text{CaO}$                       d)  $\text{PbO}$

- Q.4 Glass is a \_\_\_\_\_
- a) Super-cooled liquid    b) Solid
- c) Liquid                      d) Super-heated liquid

Q.5 Which process is used for sheet glass making?

- a) Rolling                      b) Jiggering
- c) Floating                      d) Casting

Q.6 Which of this is first step in glass manufacturing?

- a) Melting                      b) Batching
- c) Forming                      d) Annealing

### SECTION-B

**Note:** Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Stone is a type of glass defect. (True/False)
- Q.8 Silica is a glass former. (True/False)
- Q.9 Cobalt oxide gives green color to glass. (True/False)
- Q.10 Lead glass has higher density and refractive index. (True/False)

- Q.11 Tank furnace is a batch process furnace. (True/False)
- Q.12 Pot furnace is suitable for large-scale production. (True/False)

### SECTION-C

**Note:** Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Differentiate between tank furnace and pot furnace.
- Q.14 What is the role of alumina and lime in glass?
- Q.15 Explain the concept of glassy state.
- Q.16 What is the importance of chimney draft?
- Q.17 Define annealed and tempered glass with difference.
- Q.18 Write a note on oxidizing and reducing agents in glass batch.
- Q.19 Describe any one fabrication method of glass articles.
- Q.20 Discuss raw materials used in soda-lime glass.
- Q.21 Explain causes and remedies of seeds and blisters in glass.